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#include <stdio.h>
#include <stdlib.h>
struct Node {
int data;
struct Node* next;
};
struct Node* head = NULL;
// Function to create a new node struct Node* createNode(int data) {
struct Node* newNode = (struct Node*)malloc(sizeof(struct Node)); newNode->data = data;
newNode->next = NULL; return newNode;
}
// Function to insert a node at the beginning void insertAtBeginning(int data) {
struct Node* newNode = createNode(data); newNode->next = head;
head = newNode;
}
// Function to insert a node at the end void insertAtEnd(int data) {
struct Node* newNode = createNode(data); if (head == NULL) {
head = newNode; return;
}
struct Node* temp = head; while (temp->next != NULL) { temp = temp->next;
}
temp->next = newNode;
}
// Function to insert a node at a specified position void insertAtPosition(int data, int position) {
if (position < 1) { printf("Invalid position\n"); return;
}
if (position == 1) { insertAtBeginning(data); return;
}
struct Node* newNode = createNode(data); struct Node* temp = head;
int count = 1;
while (count < position - 1 && temp != NULL) { temp = temp->next;
count++;
}
if (temp == NULL) { printf("Invalid position\n"); return;
}

newNode->next = temp->next; temp->next = newNode;
}

// Function to delete a node from the beginning void deleteAtBeginning() {
if (head == NULL) { printf("List is empty\n"); return;
}
struct Node* temp = head; head = head->next; free(temp);
}

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// Function to delete a node from the end void deleteAtEnd() {

if (head == NULL) { printf("List is empty\n"); return;
}
if (head->next == NULL) { free(head);
head = NULL; return;
}
struct Node* temp = head;
while (temp->next->next != NULL) { temp = temp->next;
}
free(temp->next); temp->next = NULL;
}

// Function to delete a node at a specified position void deleteAtPosition(int position) {
if (position < 1) { printf("Invalid position\n"); return;
}
if (position == 1) { deleteAtBeginning(); return;
}
struct Node* temp = head; int count = 1;
while (count < position - 1 && temp != NULL && temp->next != NULL) { temp = temp->next;
count++;
}
if (temp == NULL || temp->next == NULL) { printf("Invalid position\n");
return;
}
struct Node* toDelete = temp->next; temp->next = toDelete->next; free(toDelete);
}

// Function to display the linked list void display() {
if (head == NULL) { printf("List is empty\n"); return;
}
struct Node* temp = head; while (temp != NULL) { printf("%d -> ", temp->data); temp =
temp->next;
}

printf("NULL\n");
}

int main() {
int choice, data, position;

while (1) {
printf("\nSingly Linked List Operations:\n"); printf("1. Insert at Beginning\n");
printf("2. Insert at End\n");
printf("3. Insert at Position\n");

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printf("4. Delete at Beginning\n");
printf("5. Delete at End\n");
printf("6. Delete at Position\n");
printf("7. Display\n");
printf("8. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
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switch (choice) { case 1:
printf("Enter data to insert at beginning: "); scanf("%d", &data); insertAtBeginning(data);
break;
case 2:
printf("Enter data to insert at end: "); scanf("%d", &data); insertAtEnd(data);
break;
case 3:
printf("Enter data to insert: ");
scanf("%d", &data);
printf("Enter position: ");
scanf("%d", &position);
insertAtPosition(data, position);
break;
case 4:
deleteAtBeginning();
break;
case 5:
deleteAtEnd();
break;
case 6:
printf("Enter position to delete: ");
scanf("%d", &position); deleteAtPosition(position);
break;
case 7:
display();
break;
case 8:
exit(0);
default:
printf("Invalid choice\n");
}
}
return 0;
}
```